THE PIONEER OF MICROBIOLOGY

# Louis Pasteur

The chemist who conquered the invisible world.

# A Mind for Observation

**1822–1895:** French chemist and microbiologist.

**Core belief:** Science must have practical applications.

**The turning point:** Investigating why wine and milk spoil.



*Louis Pasteur.*

THE EARLY YEARS

# From Tanner's Son to Dean of Sciences

### 1822: Humble Roots

Born in Dole. Initially a mediocre student preferring painting.

### 1848: First Breakthrough

Solved tartaric acid structures, launching his scientific career.

### 1854: Industrial Focus

Appointed Dean at Lille. Tasked with solving local industrial problems.

01

CORE CONCEPT TO DISCUSS

# Spontaneous Generation

The obsolete scientific theory that living organisms could arise directly and rapidly from inanimate, non-living matter.

THE PREVAILING THEORY OF THE 1800S

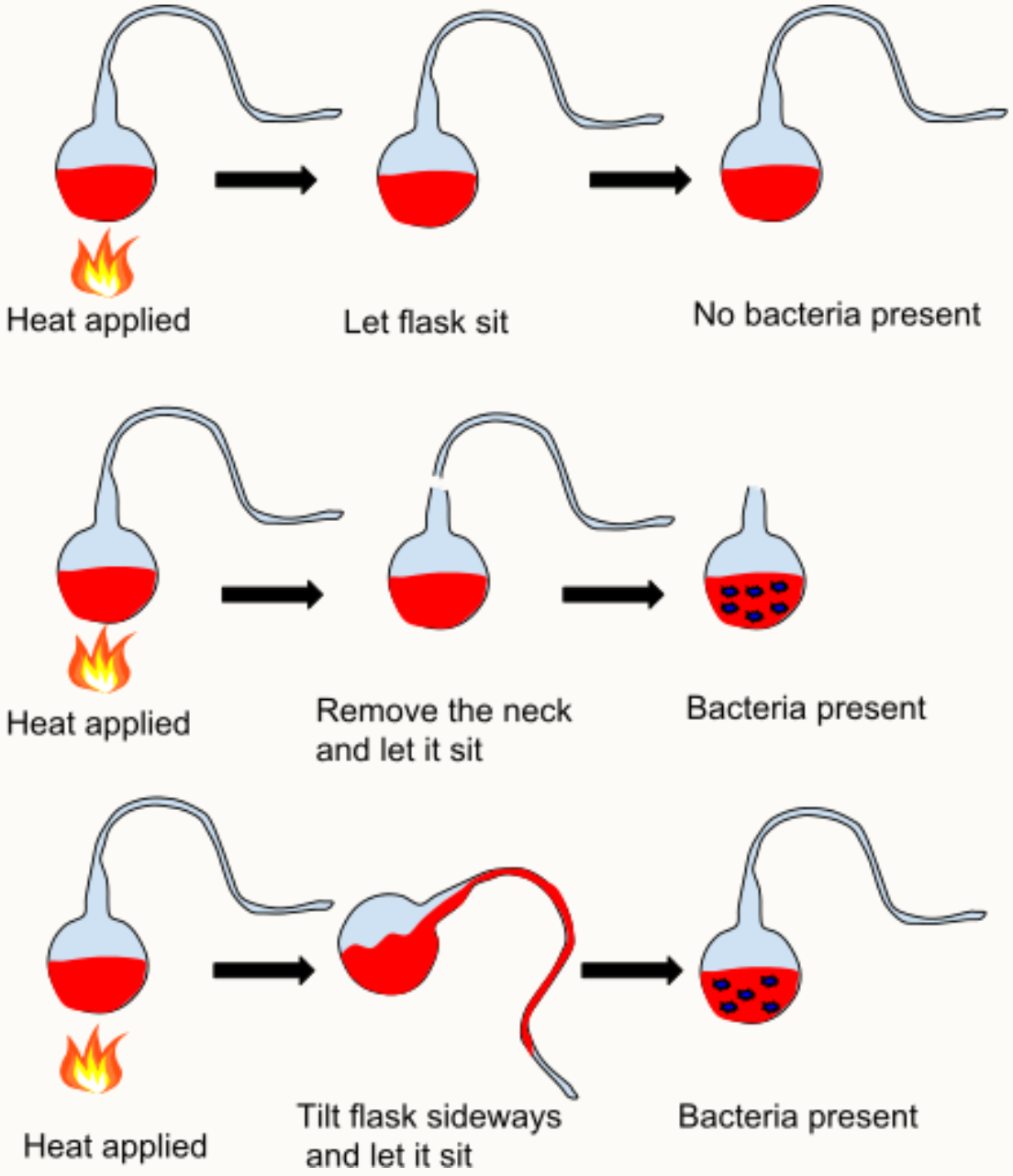
# "Spontaneous Generation"

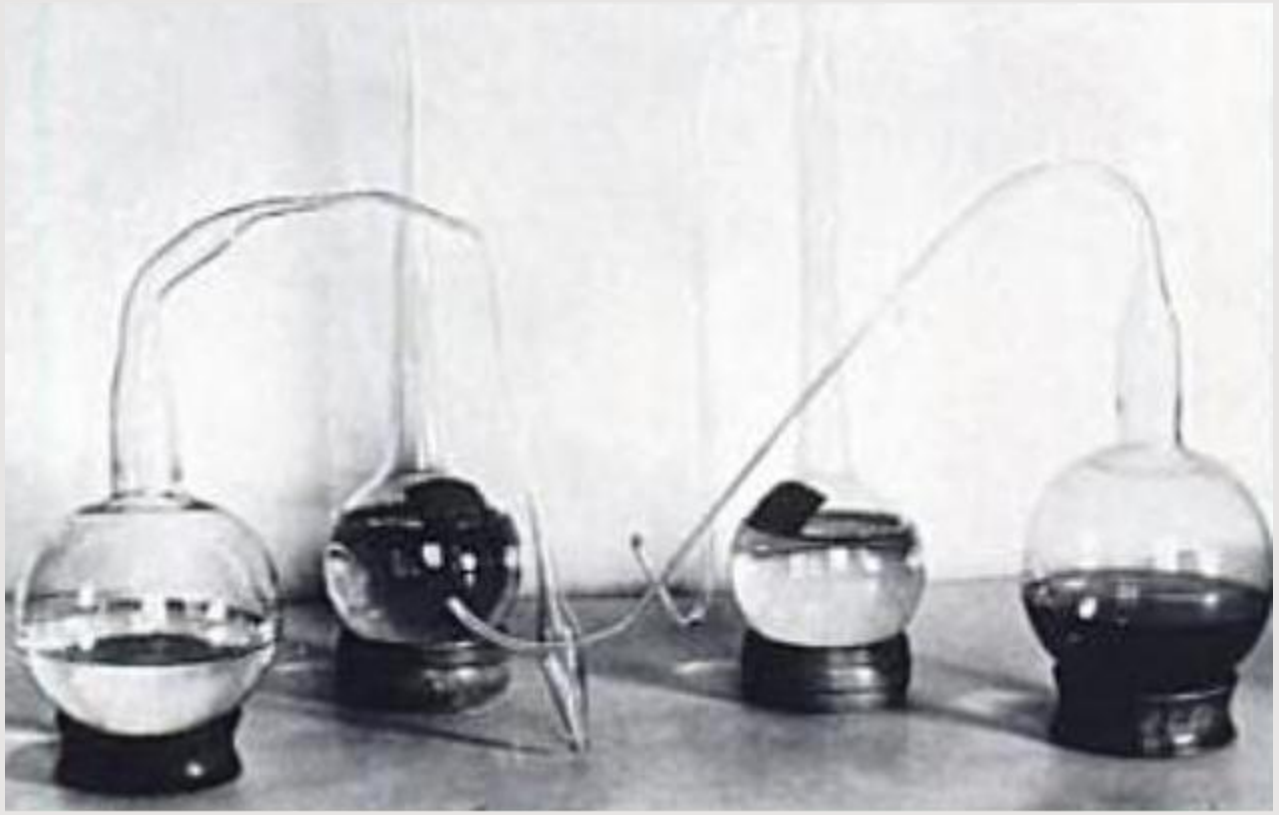
The mistaken belief that living organisms could arise directly from non-living matter.

# The Bizarre Beliefs of the Era

Before Pasteur, people believed complex life magically materialized from dead matter.

- **Rotting meat creates maggots.**
- **River mud creates frogs.**
- **Wheat and dirty rags create mice.**





# The Swan-Neck Flask

The experiment that shattered the myth.

### Step 1

Boiled broth to kill existing microbes.

### Step 2

Curved neck trapped dust and airborne particles.

### Result

Broth remained clear. Life requires existing life.

02

CORE CONCEPT TO DISCUSS

# Germ Theory

The accepted scientific theory that many diseases are caused by the presence and actions of specific microscopic organisms within the body.

THE CORE PARADIGM

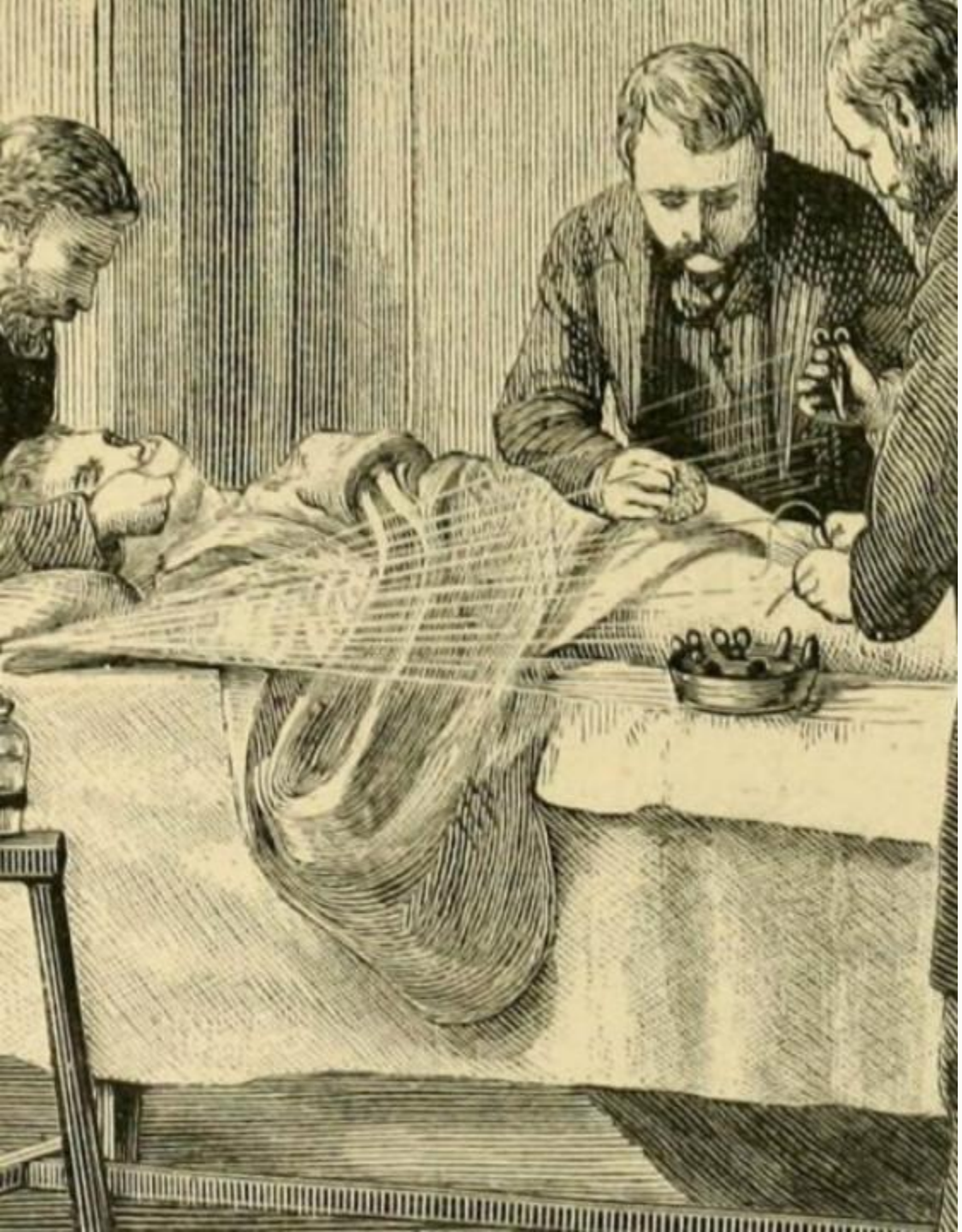
# The Germ Theory of Disease

### The Rule

Specific microscopic organisms are the true cause of specific diseases.

### The Implication

Infection is caused by external invaders, not foul air.



THE MEDICAL REVOLUTION

# Antiseptic Surgery

British surgeon Joseph Lister read Pasteur's work on microbes and realized they caused surgical infections.

- Carbolic acid used to sterilize tools.
- Surgeons forced to wash hands.
- Post-op mortality drops massively.

# The Scientific Shift

## BEFORE PASTEUR

- ✕ Spoilage just "happens" naturally.
- ✕ Disease is caused by bad air (miasma).
- ✕ No need for sterile surgical tools.

## AFTER PASTEUR

- ✓ Spoilage is caused by microbes.
- ✓ Specific germs cause specific diseases.
- ✓ Antiseptic methods become mandatory.

03

CORE CONCEPT TO DISCUSS

# Pasteurization

A process of applying mild heat to food and beverages to destroy pathogens and significantly extend shelf life without ruining the flavor.

THE ECONOMIC THREAT

# The "Diseases" of Wine

France's vital wine industry was collapsing. Vats were mysteriously turning sour and bitter.

**Pasteur's Discovery:**

He observed hostile rod-shaped bacteria contaminating the vats and producing acid.



# Saving the Wine Industry

French wine was turning to vinegar. Pasteur found the bacterial culprits and engineered a solution.

1. Gentle Heating
2. Pathogen Destruction
3. Flavor Preservation



A process still used worldwide today.

# Billions

**Of lives saved from tuberculosis and typhoid via milk pasteurization.**

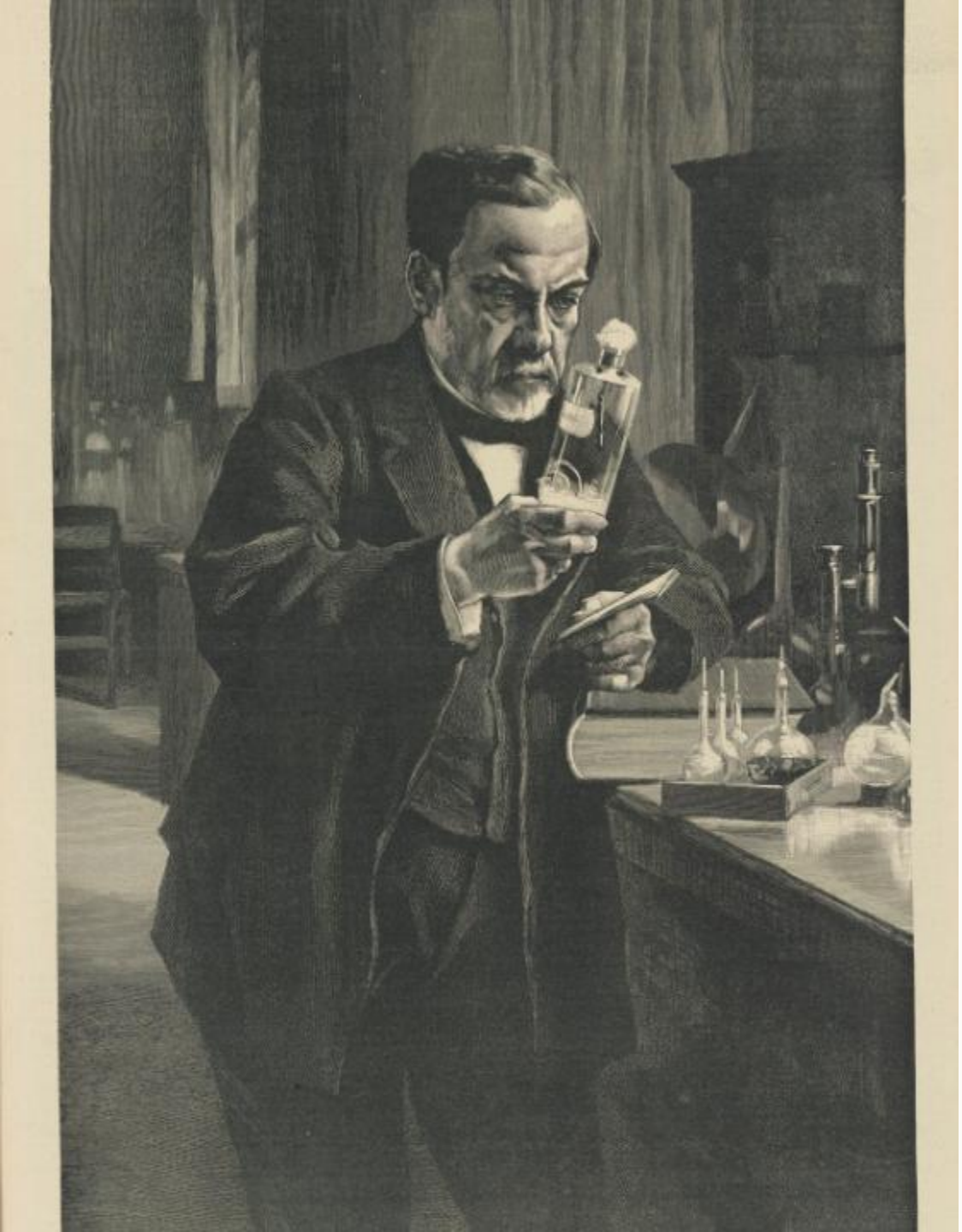
| | Pasteurization<br>HEAT TO MAKE IT SAFER | Sterilization<br>ELIMINATE ALL MICROBES |
|---|---|---|
| **Definition** | Pasteurization is the process of heating liquid to a specific temperature for a specific time period to reduce microbial growth. | Sterilization is the process of eliminating all bacterial growth from various different objects. |
| **Process** | Heat liquid to a set temperature below boiling point and then cool quickly. | Heat or chemicals used to kill all various types of bacteria. |
| **Different types** | • High-temperature, short-time (HTST)<br>• Ultra-high temperature processing (UHT) | • Applying heat<br>• Irradiation<br>• Chemicals<br>• Applying high pressure |
| **Taste** | Taste of the food remains the same. | Alters the taste of the food. |
| **Eliminates** | Some microbial bacteria that stops growth. | All various types of microbial growth including bacteria, fungi, viruses. |
| **Uses** | Liquids, Food | Food, Medical, Cleaning |

04

CORE CONCEPT TO DISCUSS

# Vaccination

The administration of a weakened or dead pathogen to stimulate an individual's immune system to develop adaptive immunity to a disease.



IMMUNOLOGY

# Taming the Microbes

Pasteur discovered that weakened pathogens could train the immune system.

### Anthrax (1881)

Protected livestock populations.

### Rabies (1885)

First successful human vaccination.

# The Rabies Gamble

**July 6, 1885**

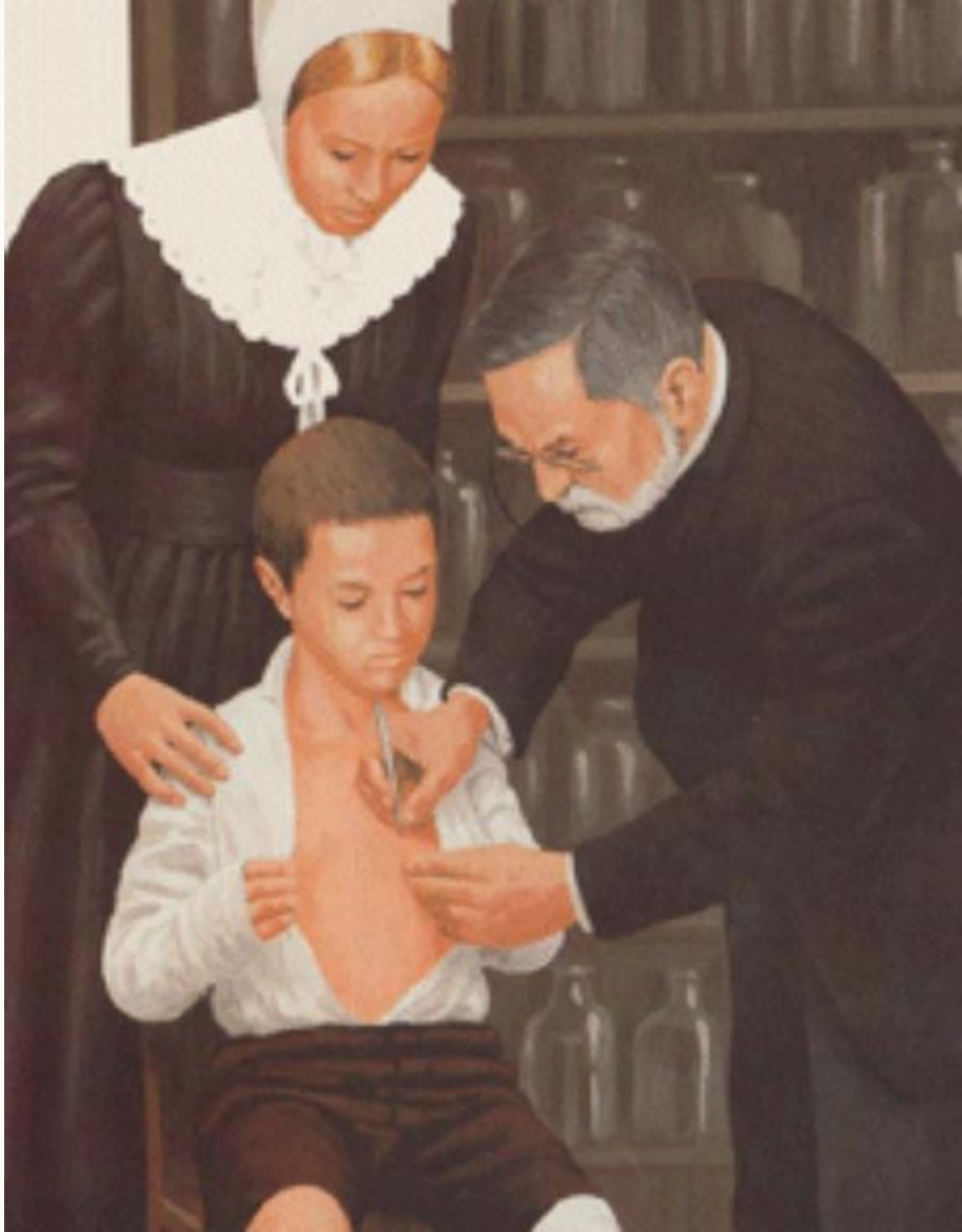
A 9-year-old boy, Joseph Meister, is severely bitten by a rabid dog. Rabies was a guaranteed, agonizing death sentence.

### The Treatment

Pasteur administers an untested, weakened rabies virus over 14 days.

### The Result

The boy never develops rabies. The era of human vaccination is born.



*"Science knows no country, because knowledge belongs to humanity, and is the torch which illuminates the world."*

— LOUIS PASTEUR

# Questions & Discussion

## How do Pasteur's discoveries impact our daily routines or lives today?

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